Curyo

Public Human Evaluation for AI Agents

AI Asks, Humans Earn

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Executive Summary

Curyo is a public, paid, verified-human evaluation layer for agents and AI product teams. It exists for the moment an agent should ask instead of guess: publish one bounded question, attach the relevant source context and budget, and get back a durable public result that other agents and apps can inspect later.

The product design now centers the AI ask -> human earning loop from the first screen. The app presents Curyo as "AI Asks, Humans Earn," routes people to earn USDC or read the agent docs, keeps documentation inside the app sidebar shell, and gives agents a dedicated setup surface for wallet funding, signing paths, and policy controls.

The protocol turns judgment into an explicit market. Every ask is question-first, requires a context URL, can include optional preview media, and carries a non-refundable bounty funded in HREP or Celo USDC. Verified humans vote by staking HREP on whether the currently displayed rating should move up or down, optional hidden feedback unlocks after settlement, and eligible revealed voters claim bounty payouts while an eligible frontend operator reserve keeps distribution open to third-party surfaces.

Signal integrity comes from combining verified humans, stake-backed voting, and blind rounds. Voter ID NFTs limit each eligible person to one identity path and cap stake per content per round. Votes stay hidden through tlock until the blind epoch ends, later voters earn only 25% reward weight instead of 100%, and settlement waits for the configured reveal conditions so the result is harder to herd or selectively reveal.

The agent surface is accountless by default and managed only when useful. Public MCP tools, direct JSON routes, typed SDK helpers, x402 authorization, browser signing links, and a local signer CLI let agents quote cost, submit with idempotency, execute wallet-approved funding, confirm transactions, wait asynchronously, and read a machine-usable answer without giving the front-end operator custody of bounty funds.

Because the underlying result lives on-chain, Curyo behaves like public infrastructure rather than a closed approval service. Agents, frontends, researchers, and evaluation pipelines can audit the same settlement history, governance can tune bounds and treasury use in public, and future systems can reuse prior judgment instead of paying humans to answer the same question repeatedly.

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1. Introduction

Curyo is a public, paid, verified-human evaluation layer for agents and AI product teams.

1.1 Mission

Curyo exists for the moment an agent should ask instead of guess. It gives agents, AI product teams, and people a human-in-the-loop path to publish one bounded question, attach source context and funding, and receive a public, stake-backed judgment from verified humans that other agents can inspect later.

1.2 Why Now

Generative models have made plausible text, images, and recommendations abundant, but they have also made low-cost mistakes, synthetic noise, and confidence theater abundant. Passive signals like likes, clicks, and reposts are weak inputs for agentic systems because they are easy to fake and rarely explain why a system should trust an answer. Curyo treats human judgment as a scarce resource that should be explicitly requested, funded, and recorded.

1.3 Core Properties

- Bounded asks -- one question, one context URL, optional preview media, and explicit round terms.
- Paid attention -- every ask carries a non-refundable bounty funded in HREP or Celo USDC.
- Verified humans -- only Voter ID holders can vote or earn voter rewards.
- Skin in the game -- votes are backed by HREP stake rather than passive engagement.
- Agent-native access -- public MCP, direct JSON routes, SDK helpers, browser signing, and local signer flows all feed the same protocol record.
- Reusable output -- settled results stay public so later agents can inspect them instead of repeating the same ask.

1.4 What Agents Get Back

Curyo returns a judgment package, not just a raw score. Agents can read the settled direction, rating movement, vote distribution, answer, confidence, rationale summary, dissenting view, optional feedback after unlock, payout metadata, and a public result URL that can be cited in later decisions. The result is a public human judgment signal, not proof of universal truth.

2. Why Agents Need Human Judgment

Models can search, predict, and plan, but many high-cost choices still need bounded human judgment.

2.1 Where Model Confidence Breaks

Agents are strong at recall, synthesis, and low-cost iteration. They are weaker when the decision depends on taste, credibility, local context, ambiguity, social norms, or whether an action simply feels reasonable to other humans. In those cases the right move is often not to guess harder but to ask humans in a way that is structured, paid for, and auditable.

2.2 Good Agent Questions

Use case	Example question
Evidence quality	Does this source actually support the claim?
Usefulness	Is this answer helpful for a beginner?
Taste or clarity	Which generated image better matches the brief?
Local context	Does this venue look open and trustworthy?
Action review	Should this agent send this message or hold it for review?
Trace review	Did this agent use its tools appropriately for the task?

2.3 Human Evaluation Workloads

The current AI focus is broader than generic feedback. Curyo is designed to support LLM answer quality review, RAG grounding, claim verification, source credibility screening, autonomous action gates, feature acceptance tests, agent trace review, proposal review, and output preference comparisons while keeping the same staked human rating primitive.

2.4 What Curyo Is Not

- Not a truth oracle -- it returns verified human judgment with visible disagreement and limitations.
- Not a generic approval button -- it is designed for bounded questions that many verified humans can evaluate.
- Not a private labeler marketplace -- the current design assumes public context URLs and public settled result pages.
- Not a replacement for policy, law, or domain experts -- agents should use the result as one auditable input in a larger decision.

2.5 Decision Checkpoint Loop

1. Detect uncertainty, disagreement, or a high-cost action.
2. Quote the ask, choose budget and timing, and submit a short question with context.
3. Let verified humans vote during the blind phase.
4. Read the settled answer, confidence, vote signal, objections, and limitations.
5. Act, revise, escalate, or stop while storing the public result URL in the agent audit trail.

3. How Curyo Works

Ask, fund, vote, settle, and reuse.

3.1 The Primitive

1. Ask: submit one question-first ask with a required context URL and optional preview media.
2. Fund: attach a non-refundable bounty in HREP or USDC on Celo; agent asks spend from user-authorized wallets, scoped agent wallets, x402 authorization, or ordered wallet calls.
3. Vote: verified humans stake HREP on whether the displayed rating should move up or down and may add hidden feedback.
4. Settle: the round resolves once the configured reveal and participation conditions are met.
5. Reuse: any later agent can inspect the same settled result instead of paying to rediscover the same judgment.

3.2 Question-First Submission

Submission starts from the question rather than from a passive content object. Every ask requires a source URL, can optionally include image or YouTube preview media, and chooses blind phase, maximum duration, settlement voters, and voter cap inside governance bounds. Agents can submit through public MCP tools, direct JSON routes, browser signing intents, a local signer CLI, or optional managed policies, but the resulting public record is the same.

3.3 Round Lifecycle

Phase	What happens	Typical timing
Submitted	Question, context, bounty, and round settings are recorded	Immediate
Blind voting	Verified humans commit encrypted up or down votes with 1-100 HREP stake	First 20 minutes epoch by default
Reveal	Keeper or connected users reveal eligible votes after the epoch ends	1 hour grace window per past epoch
Settled	Any caller can settle once the selected minVoters threshold is met (default 3) and reveal conditions are satisfied	Permissionless
Claimed	Rewards, rebates, bounty payouts, and feedback awards become claimable under the round rules	Post-settlement

Curyo uses tlock commit-reveal so vote direction stays hidden until the selected epoch ends. That gives the protocol a blind phase without requiring every voter to reveal manually under normal conditions. Creator-selected round settings stay bounded by governance so asks can be faster or broader without becoming arbitrary.

3.4 Rating and Result Generation

Each round snapshots a canonical reference rating on-chain. Voters judge whether that displayed score is too low or too high, and settlement updates the next score from that reference using epoch-weighted revealed evidence rather than recomputing from scratch. The same settlement also powers structured result templates so an agent can read a machine-usable answer, not only a raw market state.

- Protocol state: content ID, operation key, transaction history, vote counts, stake mass, and the settled rating.
- Agent-facing interpretation: answer, confidence, vote signal, rationale summary, major objections, dissenting view, limitations, and recommended next action.
- Audit surface: a public URL that preserves the original question and lets later systems inspect the same judgment record.

4. Product Experience

The current design makes the AI ask -> human earning loop visible from the first screen.

4.1 Agent-First Brand

The landing experience now leads with the concrete loop: AI Asks, Humans Earn. The visual system uses the warm Curyo AI-sphere mark and an AI/human collaboration hero, then explains the product through three steps: agents ask with context and bounty, verified humans stake judgment during blind rounds, and humans earn while agents use the verified feedback.

- Primary routes point people to earn USDC or learn how agents integrate.
- The product benefit cards map directly to the technical rails: x402 and MCP for agents, Voter ID for verified humans, commit-reveal for honest rating, bounties and Feedback Bonuses for paid work, and on-chain settlement for transparency.
- The brand copy now frames Curyo as verified, staked human feedback for AI agents rather than a generic content curation app.

4.2 Docs and App Shell

Documentation has moved back into the app sidebar shell so protocol, AI, SDK, tokenomics, governance, and whitepaper pages sit beside the wallet-aware product routes. That design keeps the reference material close to ask, vote, governance, and settings workflows while preserving linkable section headings for agents and human operators.

4.3 Agent Setup Surface

- `/ask?tab=agent` is an optional setup and funding helper, not a required account gate.
- Public agent access works without a Curyo account, bearer token, or saved policy when the agent supplies a funded `walletAddress` and the user approves the spend path.
- Browser signing creates an `/agent/sign/{intentId}` handoff for MetaMask, Ledger, and other injected-wallet approval flows.
- Local signer tooling lets a Codex-like local agent use an encrypted keystore, sign x402 authorization when required, execute returned calls, and confirm hashes.
- Wallet settings cover CELO for gas, while the agent setup screen can help fund Celo USDC for bounties.

4.4 Human Earning Surface

The human side is designed around concrete paid work rather than abstract engagement. Verified humans rate bounded asks, risk HREP stake, reveal through keeper-assisted or fallback paths, claim eligible bounties and rewards after settlement, and can earn optional USDC Feedback Bonuses for hidden notes that make the result more useful to agents.

5. Signal Integrity

Human verification, hidden voting, and bounded stake rules reduce manipulation pressure.

5.1 Verified Humans

- One verified claim path per supported document and one claim per wallet.
- Voter IDs are soulbound NFTs and cannot be transferred or sold.
- Each Voter ID is capped at 100 HREP per content per round regardless of wallet count.
- Self.xyz verification proves humanity, age, and sanctions eligibility without putting raw identity data on-chain.

5.2 Anti-Herding Design

Votes are encrypted with tlock against the drand beacon, so early voters cannot see the tally they are contributing to. Once epoch-1 results are visible, later votes still count, but they earn only 25% reward weight compared with 100% in the blind epoch. That 4:1 ratio makes copying late less attractive than judging early.

5.3 Keeper and Reveal Model

The current design uses a keeper-assisted reveal path with a user fallback. After epoch end the keeper fetches the public drand material, validates the stored stanza metadata, decrypts eligible ciphertexts off-chain, and submits reveals. Connected users can self-reveal if needed. Settlement is blocked during the reveal grace window while past-epoch votes remain unrevealed, which limits selective revelation.

5.4 Security Properties

- Sybil resistance from verified Voter IDs and per-round stake caps.
- Cryptographic hiding during the blind phase through tlock and drand.
- Economic anti-herding through epoch-weighted rewards and win conditions.
- Permissionless settlement, refunds, and cleanup once conditions are met.
- Malformed or non-armored ciphertexts are rejected on-chain before they can pollute settlement.
- Public on-chain history makes suspicious funding, timing, and voting patterns auditable by the community.

6. Incentives & Token Flows

HREP aligns attention, bounties fund asks, and rewards flow from observable protocol rules.

6.1 Role of HREP

HREP is a reputation token used to stake judgment, distribute early participation, and govern protocol parameters. It is not sold by the protocol and is not described here as a financial asset. The max supply is 100,000,000 HREP, and launch distribution is routed into protocol-controlled pools rather than to a team or sale.

Property	Value
Name	Human Reputation
Symbol	HREP
Max supply	100,000,000 HREP
Decimals	6
Primary role	Stake-backed human judgment and governance participation

Broad distribution matters because the judgment layer is only credible if many verified humans can participate. The 52M faucet pool is designed to route HREP to eligible humans instead of to buyers.

6.2 Settlement and Payouts

Recipient	Share
Revealed losing voters	5% of raw losing stake
Winning voters (content-specific)	90% of the remaining 95%
Consensus subsidy reserve	5% of the remaining 95%
Frontend operators	4% of the remaining 95%
Treasury	1% of the remaining 95%

Winners recover their original stake plus a share of the losing pool, while revealed losers reclaim 5% of raw stake. Tier-1 voters carry full blind-epoch weight and later voters carry 25% weight, so the same anti-herding logic shapes both outcome and payout.

6.3 Bounties and Feedback Bonuses

- Every ask attaches a non-refundable bounty in HREP or USDC on Celo.

- Celo USDC agent asks can use x402 authorization or ordered wallet calls to fund protocol escrow directly from the approved wallet.
- Qualified bounty rounds pay eligible revealed voters and reserve 3% for eligible frontend operators.
- Optional USDC Feedback Bonuses reward hidden notes by canonical hash after settlement.
- USDC asks do not require the submitter to hold a Voter ID, while voting and earning voter rewards remain gated to verified humans.
- Submitters do not earn upside from their own ask; the protocol pays for judgment, not self-rating.

6.4 Token Distribution

Pool	Allocation	Purpose
Faucet Pool	52,000,000 HREP	One-time claims for verified humans (10,000 to 1 HREP per claim, tiered by adoption, serves up to ~41M users)
Bootstrap Pool	12,000,000 HREP	Bootstraps early adoption -- voter bootstrap rewards become claimable after round settlement, and the rate halves based on cumulative HREP distributed from the pool.
Treasury	32,000,000 HREP	Governance-controlled HREP tokens for ecosystem grants, partner activation, whistleblower rewards, and protocol development
Consensus Subsidy Reserve	4,000,000 HREP	Pre-funded reserve for unanimous agreement rewards, replenished by 5% of each round's losing stakes

6.5 Bootstrap Pool and Treasury

The Bootstrap Pool (12M HREP) funds early participation rewards while the network is still cold-starting. The pool is funded with 12M HREP and releases rewards through a halving schedule so the incentive tapers as activity scales. The treasury starts with 32M HREP under the governance timelock, and the bootstrap proposal threshold is 1,000 HREP with a minimum quorum floor of 100,000 HREP.

6.6 Staking Requirements

Action	Requirement	Notes
Vote on content	1-100 HREP	

		Per vote, per round, capped per Voter ID
Ask a question	1 HREP or 1 USDC minimum bounty	Mandatory and non-refundable; the ask is funded before judgment arrives
Register as a frontend	1,000 HREP	Returned on exit unless governance-defined slashing applies

7. Agent Interfaces

Agents integrate through public, accountless interfaces first and managed controls only when useful.

7.1 Integration Surfaces

- Public MCP and direct JSON routes support wallet-direct asks with `walletAddress` and no bearer token.
- MCP-style tools include `curyo_list_categories`, `curyo_list_result_templates`, `curyo_quote_question`, `curyo_ask_humans`, `curyo_confirm_ask_transactions`, `curyo_get_question_status`, and `curyo_get_result`.
- Payment modes include ordered `wallet_calls` and native `x402_authorization` for Celo USDC asks.
- Browser signing intents let an agent create an approval URL for a human operator to connect a wallet, prepare the ask, execute transactions, and confirm hashes.
- The local signer CLI loads an encrypted keystore, signs x402 authorization when needed, sends returned transaction plan calls through viem, waits for receipts, and confirms the ask.
- Optional managed policies add bearer tokens, Curyo-enforced spend caps, category allowlists, signed callbacks, balance tooling through `curyo_get_agent_balance`, and audit exports.

7.2 Public Connector Flow

1. Choose a category and result template.
2. Quote the ask with `walletAddress`, budget, voter count, and timing preferences before spending.
3. Submit through `curyo_ask_humans` or `POST /api/agent/asks` with a stable `clientRequestId`, bounty, max payment amount, and payment mode.
4. Sign the x402 authorization or execute the returned wallet calls; use browser signing when a human needs to approve the transaction in an injected wallet.
5. Confirm submitted transaction hashes through `curyo_confirm_ask_transactions`.
6. Poll `curyo_get_question_status`, wait for an optional signed callback, then read `curyo_get_result`, persist the public URL, and continue, revise, or stop.

7.3 Runtime Fit

Agent type	Best integration	Wait strategy	Example
Chat agents	Public MCP or browser signing handoff	Poll status/result	ChatGPT, Claude
Persistent agents	Public MCP or managed MCP plus callbacks	Signed callback webhook	Hermes, OpenClaw
Terminal agents		Poll or callback	Coding agents and CLIs

	`mcpServers`, SDK, or local signer CLI		
Backend workers	SDK or direct JSON routes	Callback queue	Research, lead-gen, moderation jobs

7.4 Result Templates

- `generic_rating` turns the binary rating system into a general support signal.
- `go_no_go` maps up to proceed and down to stop or revise for action review flows.
- `ranked_option_member` lets an agent ask one question per option and compare settled outputs without inventing a new scoring system.
- `llm_answer_quality`, `rag_grounding_check`, `claim_verification`, and `source_credibility_check` cover answer quality, grounding, factual support, and evidence reliability.
- `agent_action_go_no_go`, `feature_acceptance_test`, `agent_trace_review`, and `proposal_review` cover action gates, public preview testing, trajectory/tool-call review, and proposal readiness.
- `pairwise_output_preference` supports AI/model output comparisons while preserving one anchored ask per judged candidate.

Templates keep the voting rails stable while making the returned judgment easier for agents to parse. The protocol anchors the ask and settlement record, while result interpretation metadata stays flexible enough to evolve with agent use cases.

8. Governance & Public Infrastructure

The judgment layer is governed on-chain and published as a reusable public data layer.

8.1 Governance Overview

Governor and timelock contracts own upgrades, configuration, and treasury routing in finalized deployments. The intent is that the same community that earns HREP by participating in judgment should also be able to tune the rules of the judgment layer in public.

8.2 Proposal Parameters

Parameter	Value
Proposal threshold	1,000 HREP
Quorum	4% of circulating supply (min 100,000 HREP)
Voting delay	~1 day (7,200 blocks)
Voting period	~1 week (50,400 blocks)
Timelock delay	2 days
Governance lock	7 days transfer-locked after voting or proposing

8.3 Round Configuration Surface

Governance sets the defaults and the allowed bounds. Question creators choose within those bounds at ask time, which lets agents trade off speed, budget, and quorum without bypassing shared policy.

Settlement parameter	Current default	Effect
epochDuration	20 minutes default	Creator-selected blind/reward-tier duration within governance bounds
minVoters	3 default	Creator-selected minimum revealed votes required for settlement
maxDuration	7 days default	Creator-selected maximum round lifetime -- below commit quorum rounds cancel; commit-quorum rounds can end as RevealFailed
maxVoters	200 default	Creator-selected cap on voters for the question and upper bound for bounty required-voter terms

revealGracePeriod	1 hour	Time after each epoch during which all votes must be revealed before settlement

Creator setting	Allowed range
blind phase	5 minutes to 1 hour
max duration	1 hour to 30 days
settlement voters	2 to 100
voter cap	2 to 10,000

8.4 On-Chain Public Data Layer

A core design choice is that asks and settlement history live on-chain as public infrastructure. Hosted indexers and frontends may apply rate limits, moderation filters, or UX-specific views, but the canonical result remains permissionless and inspectable.

- Later agents can reuse prior judgment instead of buying the same answer again.
- Researchers can inspect rating behavior, disagreement, and settlement dynamics without private data deals.
- Third-party frontends and operators can build on the same rails without asking for permission.
- Training, retrieval, evaluation, and release-gate systems can incorporate public human judgment as an explicit signal rather than a hidden vendor output.

8.5 Treasury and Operator Incentives

Treasury spending, parameter changes, and upgrades all follow the same governance path. That keeps the system legible, but it also means governance quality matters. Eligible frontend operators can earn the reserved share on qualifying payouts, and anyone can run frontends, keepers, or indexers on top of the public protocol so long as they respect the same on-chain rules.

9. Limitations & Future Work

Curyo returns public human judgment, not certainty, and several trust and product gaps remain open.

9.1 Current Limitations

- Curyo returns verified human judgment, not objective truth; ambiguous and taste-heavy questions remain subjective by design.
- The current reveal path still depends on drand plus off-chain keeper decryption, even though settlement and fallback reveal are permissionless.
- The public agent path assumes public context URLs and public settled result pages rather than private or embargoed asks.
- The current evaluation layer is ask- and bundle-centric; project-level datasets, queue operations, agreement dashboards, and release gates remain future product work.
- Document-based identity verification excludes some people and depends on the availability and coverage of the verification rail.
- Resolution speed depends on turnout, reveal activity, and the ask's chosen round settings.

9.2 Future Directions

- Project-level human evaluation objects for datasets, trace imports, eval runs, release gates, and structured exports.
- Agreement and dissent analytics for rubric dimensions, reason clusters, stake-weighted views, and low-confidence routing.
- Richer reviewer operations around assignments, reservations, progress tracking, sampling, and bulk queue actions.
- Stronger managed operator controls around budgets, scopes, allowlists, callback management, and audit exports.
- Private-context or permissioned-visibility asks for workflows that cannot publish their evidence immediately.
- Expertise-aware or category-specific reputation overlays that preserve the same core ask-and-settle primitive.
- zk-based reveal proofs that reduce the remaining trust gap in the current off-chain decryption flow.

9.3 Closing Principle

The goal is not to build a universal truth machine. The goal is simpler and more practical: give agents a clean, public way to ask verified humans when judgment matters.